

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester VI Examination, 2022-23

Paper No. CHB-605: Atomic and Molecular Structure and Spectroscopic Techniques

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer five questions in total, including Question 1 which is compulsory.***Question 1.** Answer all parts of the following: [5 × 2 = 10]

- (a) Deduce the point group and give one example of a molecule that is linear and centrosymmetric ($D_{\infty h}$).
- (b) Write the Schrödinger wave equation for the Helium atom.
- (c) State the Born-Oppenheimer approximation.
- (d) State the Beer-Lambert law.
- (e) Why is TMS (tetramethylsilane) used as an internal reference in NMR spectroscopy?

Question 2.

- (a) Draw the molecular orbital energy level diagram for the hydrogen molecule (H_2). Compare it with the Valence Bond wave function. [6]
- (b) Deduce the point group and give one example of a molecule that is non-linear and has C_{2v} symmetry. [8]

Question 3.

- (a) Explain why the $\lambda_{\max}$ of the $n \rightarrow \pi^*$ transition in an α, β -unsaturated carbonyl compound shifts to shorter wavelengths (hypsochromic shift) with increasing solvent polarity. [6]
- (b) Sketch and interpret the 1H -NMR spectra of: (i) Ethyl bromide and (ii) Acetaldehyde, showing spin-spin splitting patterns. [8]

Question 4.

- (a) Sketch the mass spectrum of 1-bromopropane, showing the characteristic isotopic pattern. How can alkyl chlorides and bromides be distinguished by mass spectrometry? [6]
- (b) Predict the expected $\lambda_{\max}$ using Woodward-Fieser rules for the following compounds: [8]
 - (i) 4-methyl-3-penten-2-one

(ii) Cholesta-3, 5-diene

Question 5. On the basis of the following spectroscopic data, suggest a structure for the compound ($\text{C}_3\text{H}_7\text{NO}$): [14]

- IR (ν_{max}) : 3428 cm^{-1} (m), $2941\text{--}2857\text{ cm}^{-1}$ (w), 1681 cm^{-1} (s), 1550 cm^{-1} (m), 1452 cm^{-1} (w).
- $^1\text{H-NMR}$ (δ , ppm) : 2.02 (s, 3H), 2.70 (s, 3H), 8.13 (s, 1H).
- $^{13}\text{C-NMR}$ (δ , ppm) : 17.7, 26.5, 171.2.